

Mengjie Zhang

Biomechanics and Ergonomics · Software Development

mengjie.zhang@auckland.ac.nz · mzha967@aucklanduni.ac.nz · mengjie-zhang.github.io

• Auckland, New Zealand



Scan for portfolio and system demos

46%

Lower ground reaction force error after deep-learning fine-tuning

84%

REBA risk-category agreement with the reference (132 trials)

4

Assessment modules in one video-based web application

2

First-author journal articles; 3 oral presentations in 2026

Education

2023 – 2027 (expected)

PhD, Biomechanics and Ergonomics

The University of Auckland, New Zealand

2020 – 2023

Master's degree, Sport Human Science

Beijing Sport University, China

2016 – 2020

Bachelor's degree, Sport Human Science

Beijing Sport University, China

Technical skills

HUMAN MOVEMENT ANALYSIS

Vicon Motion Analysis (Cortex)

OpenCap MediaPipe OpenCV

BIOMECHANICS

Inverse dynamics OpenSim

Visual3D Force platforms

Lumbar loading

ERGONOMICS

RULA REBA Revised NIOSH

3DSSPP Manual handling

SOFTWARE AND AI

Python Streamlit PyTorch

pandas Plotly MATLAB

RESEARCH AND TESTING

Validation studies SPSS

Gas analysis VO₂max Isokinetics

sEMG

Scholarship

China Scholarship Council (CSC)
doctoral scholarship, Oct 2023 – Oct 2027 (48 months)

Languages

Chinese (native) · English (professional working proficiency)

Profile

Developing validated software, grounded in biomechanics research, for human movement analysis and ergonomic design. Building on a background in sport science and sports biomechanics, my work focuses on digital human modelling, human movement analysis, occupational biomechanics and ergonomic design. Doctoral candidate at the University of Auckland, nearing thesis submission, with the PhD expected to be conferred in 2027.

Experience

Research Assistant

Jun 2026 – present

The University of Auckland, New Zealand · PI: Associate Professor Yanxin Zhang

Lead developer of a markerless system for human movement analysis from ordinary single-camera video, requiring no markers, force plates or laboratory set-up.

Gait analysis Walking kinematics, joint moments and lower-limb muscle activation	Manual-handling risk Lifting and handling tasks, including lumbar loading estimation	Rehabilitation assessment Preliminary screening and progress tracking
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- Adapted the markerless pose model, including added neck degrees of freedom, and integrated it with inverse dynamics, muscle-activation estimation and a web interface.
- Fine-tuned a deep-learning model that predicts ground reaction forces from video-derived motion, which reduced vertical ground reaction force error by 46% in lifting tasks.
- Automated exports of the analysis video, PDF report and joint-angle data.

Doctoral Researcher

2023 – present

The University of Auckland, New Zealand

- Designed and built a four-module web application that converts workplace video into postural risk, manual-handling and biomechanical outputs with automated reports.
- REBA risk categories agreed with the reference in 84% of 132 trials, and the NIOSH Lifting Index showed a mean absolute error of 0.13.
- Led a concurrent-validity study of markerless motion capture against a Vicon reference in occupational lifting (Applied Ergonomics, 2027).
- Developed a hybrid two-phase posture prediction framework for design-stage ergonomic evaluation, at approximately 85 ms per whole-body posture (WCB 2026).
- Presented at the 10th World Congress of Biomechanics and the HFESNZ 40th Conference (2026).

Experience

Core Member, Footwear Performance Testing

Mar – Oct 2022

Beijing Sport University, China

- Industry-commissioned running-economy testing of a racing-shoe series: randomised crossover study with 15 male recreational runners covering maximal oxygen uptake, running economy and lower-limb biomechanics.
- Collected data with an 8-camera infrared motion capture system (200 Hz) and a Kistler force platform (1,000 Hz); analysed outcomes with two-way repeated-measures ANOVA.

Intern, Sports Department

2021

Tsinghua University, China

- Physical fitness testing for physical education courses, including the Functional Movement Screen and body composition assessment.

Project Lead, Master's Research

Nov 2021 – Jun 2022

Beijing Sport University, China

- Designed and led a repeated-measures study of three commercially available racing shoes with different midsole bending stiffness, measuring running economy, metabolic energy cost and lower-limb joint kinematics and kinetics.
- Presented at the 40th Conference of the International Society of Biomechanics in Sports (2022).

Intern, Physical Fitness Testing for People with Disabilities

2021

Research project, Wuhan, China

- Fitness assessments including anthropometry, blood pressure, lung capacity, strength and speed.

Selected projects

Aug Max L Lower Arm
46.5°

Aug Max R Lower Arm
49.4°

⚠ Risk Summary

⚡ Risk Scores

Online video-based ergonomic assessment system

Four modules from a single standard video: RULA and REBA postural risk, the Revised NIOSH Lifting Equation, static 3D inverse dynamics and dynamic biomechanics.

84% REBA agreement

LI error 0.13

Mengjie Zhang 10TL 01.mp4

Front view · single-camera analysis

Confidence medium

Source video

3-D skeletal model

Metric 1

Model: lumbar_extension

-17.1°

ROM -42° ~ -10° range 31°

Single-camera musculoskeletal analysis

From ordinary video to 3D kinematics, predicted ground reaction forces, inverse dynamics and lower-limb muscle activation.

46% lower GRF error

Web interface

(a) Model inputs

(b) Phase I: Constrained sagittal-plane optimisation

(c) Phase II: Analytical 3D upper-limb reconstruction

(d) Predicted posture

Hybrid two-phase posture prediction

Predicts working posture from task specifications before a workstation exists, for design-stage ergonomic evaluation.

≈ 85 ms per posture

WCB 2026

Footwear performance testing

Running economy and lower-limb biomechanics of racing shoes with different midsole bending stiffness.

15 runners

ISBS 2022

Selected publications and presentations

[1]

Zhang, M., Nieuwenhuys, A., Boocock, M.* & Zhang, Y.* (2027). Concurrent validity of markerless motion capture for occupational lifting: Kinematics and ergonomic risk assessment. *Applied Ergonomics*, 138, 104866.

[2]

Zhang, M., Nieuwenhuys, A., & Zhang, Y.* (2025). Posture prediction models in digital human modeling for ergonomic design: A systematic review. *Medical Engineering & Physics*, 143, 104391.

[3]

Oral presentations: 10th World Congress of Biomechanics, Vancouver (2 presentations, 2026) · HFESNZ 40th Conference, Wellington (2026) · 2nd Human Sports Science Conference (2022).

* Corresponding author.

Volunteering

Beijing Marathon (2017) · Beijing Tongzhou Marathon (2021) · Beijing Sport University COVID-19 vaccination programme (2021)